

Tugas Praktikum Software Engineering Pertemuan 10

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CI/CD dan DevOps Menggunakan GitHub Actions

1. Repository Github

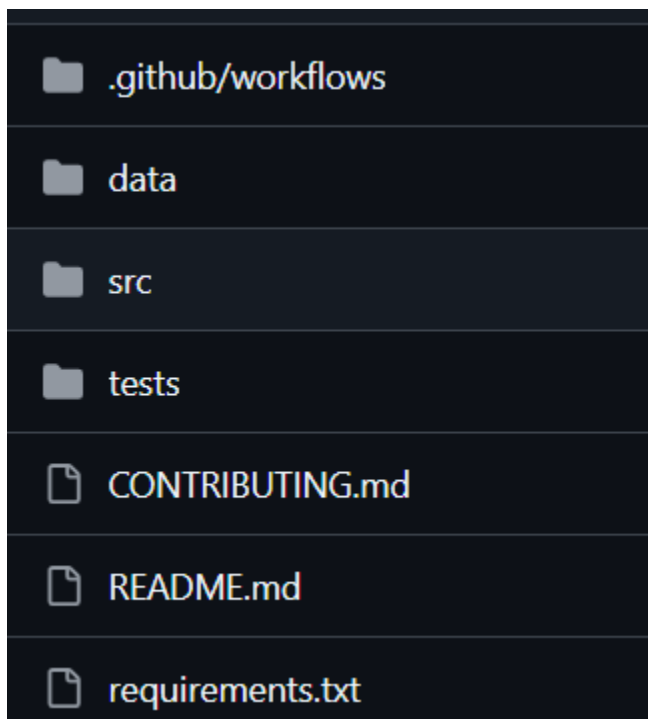
[Link Repository](#)

[Link Branch](#)

[Link Pull Request](#)

2. Struktur Repository

Struktur repository yang digunakan pada praktikum ini terdiri dari beberapa folder dan file utama sebagai berikut:



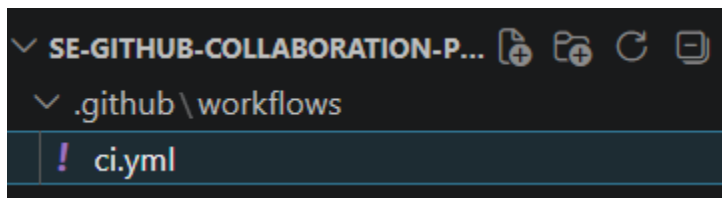
Penjelasan:

- [.github/workflows/](#) : Menyimpan workflow GitHub Actions.

- `src/` : Berisi source code aplikasi.
- `tests/` : Berisi unit testing aplikasi.
- `data/` : Menyimpan data yang digunakan aplikasi.
- `requirements.txt` : Daftar dependency Python.
- `README.md` : Dokumentasi proyek.
- `CONTRIBUTING.md` : Panduan kontribusi.

3. File Workflow CI/CD

Workflow CI/CD dibuat menggunakan GitHub Actions dengan file konfigurasi `ci.yml` yang disimpan pada folder `.github/workflows`.



Langkah-langkah Implementasi

a. Membuat Branch Fitur

Membuat branch baru untuk pengembangan pipeline CI/CD.

```
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git pull origin main
From https://github.com/AnisaIntania/se-github-collaboration-practice-
 * branch          main          -> FETCH_HEAD
Already up to date.
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git checkout -b feature/ci-pipeline-teesa
Switched to a new branch 'feature/ci-pipeline-teesa'
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git branch
feature-validasi-status-teesa
* feature/ci-pipeline-teesa
  fitur-tambah-task
  main
```

b. Menjalankan Unit Testing

Pengujian dilakukan menggunakan Pytest untuk memastikan seluruh fungsi berjalan dengan baik.

Hasil pengujian menunjukkan: 5 passed

yang berarti seluruh test berhasil dijalankan tanpa error.

```
● PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> python -m pytest
===== test session starts =====
platform win32 -- Python 3.13.13, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\Teesa\Downloads\se-github-collaboration-practice-
collected 5 items

tests\test_task_manager.py ..... [100%]

===== 5 passed in 0.08s =====
● PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git status
On branch feature/ci-pipeline-teesa
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/_pycache_/task_manager.cpython-313.pyc
        modified:   src/task_manager.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .github/
        requirements.txt
        tests/

no changes added to commit (use "git add" and/or "git commit -a")
```

c. Commit dan Push Perubahan

Perubahan yang telah dibuat kemudian di-commit dan dikirim ke repository GitHub.

```
● >> git commit -m "feat: add CI pipeline with automated tests"
[feature/ci-pipeline-teesa b7f41c7] feat: add CI pipeline with automated tests
6 files changed, 130 insertions(+), 61 deletions(-)
create mode 100644 .github/workflows/ci.yml
create mode 100644 requirements.txt
create mode 100644 tests/_pycache_/test_task_manager.cpython-313-pytest-9.0.3.pyc
create mode 100644 tests/test_task_manager.py
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git push origin feature/ci-pipeline-teesa
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (14/14), 5.10 KiB | 579.00 KiB/s, done.
Total 14 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/ci-pipeline-teesa' on GitHub by visiting:
remote:   https://github.com/AnisaIntania/se-github-collaboration-practice-/pull/new/feature/ci-pipeline-teesa
remote:
To https://github.com/AnisaIntania/se-github-collaboration-practice-.git
* [new branch]   feature/ci-pipeline-teesa -> feature/ci-pipeline-teesa
```

```

PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git add .
>> git commit -m "fix: resolve src module import issue"
>> git push
[feature/ci-pipeline-teesa 49699c9] fix: resolve src module import issue
 2 files changed, 1 insertion(+), 1 deletion(-)
   create mode 100644 src/__init__.py
fatal: The current branch feature/ci-pipeline-teesa has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin feature/ci-pipeline-teesa

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git push --set-upstream origin feature/ci-pipeline-teesa
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 618 bytes | 154.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/AnisaIntania/se-github-collaboration-practice-.git
   b7f41c7..49699c9  feature/ci-pipeline-teesa -> feature/ci-pipeline-teesa
branch 'feature/ci-pipeline-teesa' set up to track 'origin/feature/ci-pipeline-teesa'.

```

```

PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git add .
>> git commit -m "test: simulate failed pipeline"
>> git push
Enumerating objects: 45, done.
Counting objects: 100% (45/45), done.
Delta compression using up to 12 threads
Compressing objects: 100% (38/38), done.
Writing objects: 100% (45/45), done.
Total 45 (delta 12), reused 0 (delta 0), pack-reused 0 (from 0)
[feature/ci-pipeline-teesa 7c94003] test: simulate failed pipeline
 1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 403 bytes | 403.00 KiB/s, done.
Total 4 (delta 2), reused 2 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/AnisaIntania/se-github-collaboration-practice-.git
   49699c9..7c94003  feature/ci-pipeline-teesa -> feature/ci-pipeline-teesa
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice->

```

```

PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> git add .
>> git commit -m "fix: repair failing test"
>> git push
[feature/ci-pipeline-teesa c258fee] fix: repair failing test
 1 file changed, 1 insertion(+), 1 deletion(-)
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 402 bytes | 402.00 KiB/s, done.
Total 4 (delta 2), reused 1 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/AnisaIntania/se-github-collaboration-practice-.git
 6b070f1..c258fee  feature/ci-pipeline-teesa -> feature/ci-pipeline-teesa
PS C:\Users\Teesa\Downloads\se-github-collaboration-practice-> 

```

4. Hasil Workflow GitHub Actions

Setelah perubahan di-push ke repository, GitHub Actions secara otomatis menjalankan pipeline CI/CD.

Hasil workflow menunjukkan beberapa percobaan yang gagal pada tahap awal, kemudian berhasil setelah dilakukan perbaikan pada source code dan unit test.

Status workflow:

- ❌ Pipeline gagal saat simulasi error.
- ❌ Pipeline gagal karena masalah import module.
- ✅ Pipeline berhasil setelah perbaikan test dan source code.

5 workflow runs		Workflow ▾	Event ▾	Status ▾	Branch ▾	Actor ▾
✅	fix: repair failing test	feature/ci-pipeline-teesa		28 minutes ago 14s	...	
CI Pipeline #5: Commit c258fee pushed by shzsaa						
❌	fix: repair failing test	feature/ci-pipeline-teesa		30 minutes ago 22s	...	
CI Pipeline #4: Commit 6b070f1 pushed by shzsaa						
❌	test: simulate failed pipeline	feature/ci-pipeline-teesa		31 minutes ago 14s	...	
CI Pipeline #3: Commit 7c94003 pushed by shzsaa						
✅	fix: resolve src module import issue	feature/ci-pipeline-teesa		33 minutes ago 23s	...	
CI Pipeline #2: Commit 49699c9 pushed by shzsaa						
❌	feat: add CI pipeline with automated tests	feature/ci-pipeline-teesa		39 minutes ago 27s	...	
CI Pipeline #1: Commit b7f41c7 pushed by shzsaa						

5. kesimpulan

Pada praktikum ini berhasil diterapkan proses Continuous Integration (CI) menggunakan GitHub Actions. Setiap perubahan kode yang di-push ke repository akan secara otomatis menjalankan proses testing sehingga kesalahan dapat dideteksi lebih cepat. Implementasi CI/CD membantu meningkatkan kualitas perangkat lunak, mempercepat proses pengembangan, serta meminimalkan risiko error saat integrasi kode.